

Maximum measure of zero-sum-free sets on high-dimensional spheres

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Abstract

Let σ_{d-1} denote normalized surface measure on the unit sphere $S^{d-1} \subseteq \mathbb{R}^d$. We prove that, for every $d \geq 3$, any measurable set $A \subseteq S^{d-1}$ containing no triple $(x, y, z) \in A^3$ with $x + y + z = 0$ satisfies $\sigma_{d-1}(A) \leq 1/2$. This bound is sharp, as open hemispheres show, and answers a question of Bukh.

1 Introduction

Motivated by a question on Bukh's problem page [2], we consider measurable subsets of the unit sphere with no zero-sum triples. More precisely, let $d \geq 3$, and let $A \subseteq S^{d-1}$ be a measurable set containing no triple satisfying

$$x + y + z = 0, \quad x, y, z \in A. \quad (1)$$

Must A have normalized surface measure at most $1/2$?

An open hemisphere gives the matching lower bound. Indeed, let

$$H := \{(x_1, \dots, x_d) \in S^{d-1} : x_1 > 0\}.$$

Then H contains no triple satisfying (1), and $\sigma_{d-1}(H) = 1/2$.

We show that this example is extremal in all dimensions $d \geq 3$. For $d \geq 2$, define

$$\alpha_d = \sup \left\{ \sigma_{d-1}(A) : \begin{array}{l} A \subseteq S^{d-1} \text{ is measurable, and} \\ A \text{ contains no triple satisfying (1)} \end{array} \right\}.$$

The main result is the following sharp bound.

Theorem 1.1. *For every $d \geq 3$, $\alpha_d = \frac{1}{2}$.*

The restriction $d \geq 3$ is essential. In contrast, on S^1 the zero-sum triples are exactly the orbits of the rotation by $2\pi/3$. Indeed, if $x + y + z = 0$ and $x, y, z \in S^1$, then $\|x + y\|^2 = \|-z\|^2 = 1$, so $x \cdot y = -1/2$, and similarly for the other pairs.

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Identify S^1 with $\mathbb{R}/\mathbb{Z}$, and let $R(t) = t + 1/3$. If $A \subseteq S^1$ is measurable and contains no zero-sum triple, then for almost every $t \in [0, 1/3)$,

$$\sum_{j=0}^2 \mathbf{1}_A(t + j/3) \leq 2.$$

Integrating over $[0, 1/3)$ gives $\sigma_1(A) \leq 2/3$. Conversely, an open arc of normalized length $2/3$ contains no complete orbit of R , and so contains no zero-sum triple. Hence $\alpha_2 = 2/3$.

2 The proof

The proof uses, with some elementary arguments, the case $k = 3$ of Bollobás's result on spherical k -diameters [1], in the formulation quoted by Erdős, Hajnal, Simonovits, Sós, and Szemerédi [3, Definition 4.4 and the Bollobás theorem following it, p. 315]. We record the needed form below.

We use Euclidean distance on the sphere. If E is a subset of a metric space and $k \geq 2$, define the k -diameter of E by

$$d_k(E) = \sup_{x_1, \dots, x_k \in E} \min_{1 \leq i < j \leq k} \text{dist}(x_i, x_j).$$

In this note, $\text{dist}(x, y) = \|x - y\|_2$.

Theorem 2.1 (Bollobás [1]). *Let E be a nonempty subset of a sphere S^h , endowed with the metric inherited from the ambient Euclidean space, with normalized outer surface measure $\mu^*(E)$, and let C be a spherical cap whose normalized surface measure is $\mu^*(E)$. Then $d_3(E) \geq d_3(C)$.*

The theorem is stated for outer measure; in the proof below we apply it only to a compact set, for which outer measure and measure agree. We apply the theorem with the metric inherited from the ambient Euclidean space. Equivalently, if the theorem is stated using angular distance θ , the same comparison follows for Euclidean distance because $\|x - y\|_2 = 2 \sin(\theta(x, y)/2)$ is an increasing function of $\theta(x, y) \in [0, \pi]$.

Lemma 2.2. *For any three points $x, y, z \in S^{d-1}$,*

$$\min\{\|x - y\|, \|y - z\|, \|z - x\|\} \leq \sqrt{3}.$$

Moreover, if all three distances are at least $\sqrt{3}$, then

$$x \cdot y = x \cdot z = y \cdot z = -\frac{1}{2}, \quad x + y + z = 0.$$

Proof. The condition $\|x - y\| \geq \sqrt{3}$ is equivalent to $x \cdot y \leq -1/2$. Indeed, since $x, y \in S^{d-1}$,

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2x \cdot y = 2 - 2x \cdot y.$$

If all three pairwise distances are at least $\sqrt{3}$, then

$$\|x + y + z\|^2 = 3 + 2(x \cdot y + x \cdot z + y \cdot z) \leq 3 + 2 \cdot 3 \left(-\frac{1}{2}\right) = 0.$$

Thus equality holds throughout. This gives $x \cdot y = x \cdot z = y \cdot z = -1/2$ and $x + y + z = 0$. In particular, it is impossible for all three pairwise distances to be strictly larger than $\sqrt{3}$, which proves the first assertion. $\square$

Proof of Theorem 1.1. The open hemisphere example gives $\alpha_d \geq 1/2$. We prove the reverse inequality.

Assume, for contradiction, that $A \subseteq S^{d-1}$ is measurable, contains no triple satisfying (1), and has $\sigma_{d-1}(A) > 1/2$. By regularity of surface measure on the compact metric space S^{d-1} , choose a compact set $F \subseteq A$ with $\sigma_{d-1}(F) > 1/2$.

Let C be a spherical cap with the same normalized surface measure as F . We may write

$$C = \{u \in S^{d-1} : u \cdot a \geq t\}$$

for some unit vector a and some $t \in [-1, 1]$. Since $\sigma_{d-1}(C) > 1/2$, we have $t < 0$; hence C contains the equator

$$\{u \in S^{d-1} : u \cdot a = 0\} \cong S^{d-2}.$$

Because $d \geq 3$, this equator contains a circle, and hence contains three points on a common great circle whose angular separations are 120° . The Euclidean distances between these three points are all $\sqrt{3}$. By Lemma 2.2, no larger value of d_3 is possible on the sphere. Therefore

$$d_3(C) = \sqrt{3}.$$

By Theorem 2.1,

$$d_3(F) \geq d_3(C) = \sqrt{3}.$$

Since F is compact and the function

$$(x, y, z) \mapsto \min\{\|x - y\|, \|y - z\|, \|z - x\|\}$$

is continuous, the supremum in the definition of $d_3(F)$ is attained. Thus there are points $x, y, z \in F$ such that all three pairwise distances are at least $\sqrt{3}$. Lemma 2.2 then gives $x + y + z = 0$. Since $F \subseteq A$, this contradicts the assumption on A .

Therefore every measurable zero-sum-free set A has measure at most $1/2$, and so $\alpha_d = 1/2$. $\square$

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Declaration on the use of AI

The authors used generative AI tools to assist with discussing proof strategies, proof checking, and exposition. All mathematical arguments, results, and conclusions were reviewed and verified by the authors.

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